Autonomous Industrial IoT Digital Twin & Predictive Maintenance Engine with Closed-Loop ERP Action

SAP Technologies Capstone Project — Detailed Project Document

1. Executive Summary

This project designs and develops an Autonomous Industrial IoT Digital Twin and Predictive Maintenance Engine that monitors critical industrial equipment in real time, predicts impending failures using machine learning and deep learning, and automatically initiates maintenance and inventory workflows in SAP S/4HANA.

Unlike conventional predictive-maintenance projects that stop at anomaly detection or dashboard visualization, this system implements a closed-loop enterprise workflow: physical sensor anomaly → AI diagnosis → RUL prediction → agentic decision → SAP BTP → SAP PM/MM action → SAP Fiori visibility.

The system collects vibration, temperature, and motor-current data from an ESP32-based edge device. Telemetry is transmitted through MQTT to AWS IoT Core, processed by AWS Lambda, stored in Amazon Timestream/DynamoDB and Amazon S3, and analyzed using FFT-based feature engineering and deep-learning models.

When the system detects abnormal vibration, overheating, abnormal current draw, or declining Remaining Useful Life (RUL), it autonomously diagnoses the likely failure mode, checks spare-part availability in SAP MM, evaluates maintenance urgency, creates a SAP PM maintenance notification/work order, triggers procurement if required, and displays the complete status in SAP Fiori. SAP provides OData-based Maintenance Notification APIs for creating and processing maintenance notifications. [web:67][web:68]

2. Problem Statement

Industrial equipment such as motors, pumps, compressors, conveyors, gearboxes, and turbines is central to manufacturing operations. Unexpected failure can cause production stoppages, safety risks, quality defects, emergency repair costs, and supply-chain disruption.

Traditional maintenance approaches are limited: reactive maintenance repairs after failure; calendar-based preventive maintenance may replace healthy parts too early or miss early failures; condition-based maintenance improves monitoring but may still lack prediction; predictive maintenance uses sensor data and AI to predict failures before they occur.

The core problem is that many organizations collect equipment data but fail to convert it into timely, automated business action. A maintenance engineer may see an alert, but work-order creation, spare-part checking, procurement, and scheduling may remain manual, delayed, or disconnected.

This project addresses that gap by building an autonomous pipeline from physical sensor anomaly to SAP ERP action.

3. Project Objectives

3.1 Primary Objective

To design, develop, and demonstrate an end-to-end autonomous predictive-maintenance system that monitors industrial equipment using IoT sensors, predicts failures using deep learning, and automatically initiates SAP S/4HANA Plant Maintenance and Materials Management workflows through SAP BTP.

3.2 Specific Objectives

• Develop an ESP32-based IoT edge device to collect vibration, temperature, and motor-current data.

• Transmit telemetry securely to AWS IoT Core using MQTT.

• Build a serverless AWS data pipeline using AWS Lambda, Amazon Timestream/DynamoDB, and Amazon S3.

• Create a digital twin representing live asset condition, health score, anomaly state, and predicted maintenance requirement.

• Apply time-domain and FFT-based frequency-domain feature engineering to raw sensor data.

• Develop an LSTM Autoencoder for unsupervised anomaly detection.

• Develop a 1D-CNN model for supervised fault classification.

• Develop a CNN-LSTM regression model for Remaining Useful Life prediction.

• Implement an Isolation Forest baseline for comparison.

• Implement MLOps using DVC, MLflow, and Evidently AI.

• Build Diagnostic, Inventory, and Scheduler agents using LangGraph or CrewAI.

• Integrate the AI engine with SAP S/4HANA through SAP BTP, SAP Integration Suite, and OData APIs.

• Automatically create SAP PM maintenance notifications/work orders when risk thresholds are exceeded.

• Automatically verify spare-part stock and lead times through SAP MM.

• Trigger SAP MM purchase requisitions when spare parts are unavailable.

• Build an SAP Fiori dashboard for asset health, alerts, predictions, work orders, and inventory status.

• Evaluate model performance, system latency, SAP integration reliability, and business impact.

4. Project Scope

4.1 In Scope

The project covers one critical rotating asset, such as a DC motor, pump, compressor, or conveyor drive. It includes ESP32-based sensor-node development or Wokwi simulation; vibration, temperature, and current monitoring; MQTT telemetry; AWS IoT Core ingestion; AWS Lambda processing; time-series and object storage; FFT and statistical feature extraction; anomaly detection and RUL prediction; MLOps; agentic AI; SAP BTP integration; SAP PM work-order automation; SAP MM inventory and procurement automation; SAP Fiori dashboard; and testing/documentation.

4.2 Out of Scope

Factory-wide deployment, PLC/SCADA/OPC-UA integration beyond the prototype, production SAP implementation and licensing, automatic physical repair or machine shutdown, safety-critical control, full procurement execution, enterprise cybersecurity certification, and training on confidential industrial datasets are excluded.

5. Business Value

5.1 Operational Benefits

Early detection of bearing wear, rotor imbalance, misalignment, overheating, and abnormal motor load; reduced unplanned downtime; improved maintenance planning; reduced emergency repair costs; better technician prioritization; improved spare-part availability; increased asset availability and reliability.

5.2 SAP Business Benefits

Automated SAP PM work-order creation, risk-based maintenance prioritization, automated spare-part verification, reduced manual data entry, better integration between operational technology and ERP, improved auditability, and faster maintenance response.

5.3 Strategic Benefits

Demonstrates Industry 4.0 maturity, connects OT, cloud AI, and ERP, provides a reusable architecture for multiple assets, creates a foundation for autonomous maintenance operations, and supports sustainability by avoiding unnecessary part replacement and reducing energy waste.

6. System Architecture

The system follows an event-driven architecture in which sensor telemetry is ingested through AWS, analyzed by AI, evaluated by agents, and converted into SAP PM/MM business actions through SAP BTP and OData APIs.

```
Physical Asset
↓
ESP32 + MPU-6050 + DS18B20/DHT22 + ACS712
↓ MQTT over TLS
AWS IoT Core → IoT Rules → AWS Lambda
↓ ↓
Amazon Timestream/DynamoDB Amazon S3
↓ ↓
Feature Engineering: time-domain + FFT features
↓
AI Models: Isolation Forest, LSTM Autoencoder,
1D-CNN classifier, CNN-LSTM RUL regressor
↓
MLOps: DVC + MLflow + Evidently AI
↓
Agentic AI: Diagnostic, Inventory, Scheduler Agents
↓ OData v4 / REST
SAP BTP Integration Suite / API Management
↓ ↓
SAP S/4HANA PM SAP S/4HANA MM
↓ ↓
Maintenance work order Stock check / purchase requisition
↓ ↓
SAP Fiori Dashboard
```

7. Hardware Architecture

7.1 Hardware Components

Component	Specification	Purpose
ESP32	30-pin Wi-Fi/Bluetooth board	Main edge controller
MPU-6050	3-axis accelerometer + gyroscope	Vibration and motion monitoring
DS18B20 / DHT22	Digital temperature sensor	Bearing or motor temperature
ACS712	Hall-effect current sensor	Motor current/load monitoring
DC motor	5V–12V with unbalanced rotor	Generates realistic vibration signatures
Prototyping kit	Breadboard, wires, 5V 2A adapter	Circuit assembly and power
Wokwi	Online ESP32 simulator	Zero-hardware fallback

7.2 Sensor Placement and Edge Processing

The MPU-6050 is mounted on the motor body or bearing housing; DS18B20 is placed near the bearing or winding area; ACS712 is placed in series with the motor power line. The ESP32 samples sensors, applies moving-average filtering, removes outliers, calculates RMS, adds timestamps and IDs, creates JSON, and publishes MQTT telemetry.

```
{
"device_id": "ESP32-MOTOR-001",
"asset_id": "MOTOR-001",
"timestamp": "2026-10-09T19:45:00Z",
"accel_x": 0.42, "accel_y": -0.18, "accel_z": 9.71,
"temperature_c": 54.8, "current_a": 1.32,
"rms_vibration": 0.61
}
```

If physical hardware is unavailable, Wokwi can simulate the ESP32 and sensors, while NASA C-MAPSS provides multivariate run-to-failure time-series data with RUL labels for prognostic-model development. [web:65]

8. Cloud Architecture

8.1 AWS IoT Core

AWS IoT Core is the secure entry point for device telemetry. It provides device identity/authentication, MQTT brokering, topic-based routing, IoT Rules, and integration with Lambda, S3, DynamoDB, and Timestream.

```
factory/plant01/motor001/telemetry
factory/plant01/motor001/alert
factory/plant01/motor001/health
factory/plant01/motor001/command
```

8.2 AWS Lambda Functions

Lambda function	Responsibility
telemetry_ingest_lambda	Validates and enriches incoming MQTT messages
feature_engineering_lambda	Computes RMS, FFT, kurtosis, and other features
inference_lambda	Runs anomaly detection and RUL prediction
alert_lambda	Creates alerts and invokes SAP workflow
sap_integration_lambda	Prepares OData payload and calls SAP BTP
retraining_trigger_lambda	Starts retraining when drift is detected

8.3 Data Storage Strategy

Data type	Storage service	Purpose
Raw telemetry	Amazon S3	Long-term storage and training data
Live/recent telemetry	Amazon Timestream or DynamoDB	Dashboard queries and monitoring
Asset metadata	Amazon DynamoDB	Asset registry and configuration
Feature datasets	Amazon S3	Model training and reproducibility
Model artifacts	Amazon S3	Versioned model weights
Alerts/events	DynamoDB or SNS	Alert history and notifications

9. Digital Twin Design

A digital twin is a virtual representation of the physical asset that reflects its current state, historical behavior, and predicted future condition.

9.1 Digital Twin Data Model

```
{
"asset_id": "MOTOR-001",
"asset_name": "Conveyor Drive Motor",
"plant": "PLANT-01",
"criticality": "HIGH",
"current_status": "RUNNING",
"health_score": 72,
"anomaly_score": 0.81,
"risk_level": "HIGH",
"predicted_failure_mode": "Bearing Wear",
"rul_hours": 38,
"temperature_c": 61.4,
"current_a": 1.58,
"rms_vibration": 0.94,
"sap_equipment_number": "10000021",
"active_work_order": "4000123"
}
```

9.2 Asset States

State	Description
Healthy	Normal vibration, temperature, and current behavior
Watchlist	Minor deviation; monitoring frequency increases

State	Description
Warning	Anomaly detected; maintenance should be planned
Critical	High failure probability; urgent maintenance required
Maintenance	SAP PM work order created; asset under maintenance
Offline	No telemetry received

9.3 Health Score

A composite health score can be calculated as: HealthScore = 100 - (w1*A + w2*T + w3*C + w4*R), where A is normalized anomaly score, T is temperature risk, C is current risk, R is RUL risk, and w1–w4 are configurable weights.

10. Data Pipeline and Feature Engineering

10.1 Data Flow

```
ESP32 sensor readings
↓ MQTT
AWS IoT Core
↓ IoT Rule
AWS Lambda
↓
Validation and enrichment
↓
Feature extraction
↓
Model inference
↓
Alert evaluation
↓
SAP BTP integration
↓
SAP PM/MM action
↓
Fiori dashboard
```

10.2 Time-Domain Features

Feature	Maintenance meaning
RMS	Overall vibration energy
Peak / peak-to-peak	Sudden shocks or impacts
Kurtosis	Impulsiveness; often rises with bearing defects
Crest factor	Peak severity relative to RMS
Skewness	Asymmetry in vibration distribution
Standard deviation	Operational stability
Impulse factor	Impact-like fault behavior

10.3 Frequency-Domain Features

FFT converts time-domain vibration into a frequency spectrum, making fault-related frequency peaks easier to identify. Bearing, imbalance, misalignment, and gear faults create distinct frequency-domain signatures.

Feature	Meaning
Dominant frequency	Strongest vibration component
Spectral centroid	Center of frequency energy
Spectral spread	Frequency dispersion
Band energy	Energy in selected frequency ranges

Feature	Meaning
Top FFT peaks	Major fault-related frequencies
Harmonic energy	Energy at multiples of fundamental frequency
Spectral entropy	Complexity/randomness of spectrum

For bearing analysis, envelope-spectrum features related to BPFO, BPFI, BSF, and FTF frequencies can also be extracted.

11. Machine Learning Models

11.1 Model Portfolio

Model	Learning type	Purpose	Output
Isolation Forest	Unsupervised	Baseline anomaly detection	Anomaly score, binary label
LSTM Autoencoder	Unsupervised	Main anomaly detection	Reconstruction error, anomaly score
1D-CNN	Supervised	Fault classification	Fault class and probability
CNN-LSTM Regressor	Supervised regression	RUL prediction	Remaining useful life
Rule Engine + Agents	Decision logic	Converts predictions to action	SAP PM/MM action

11.2 Isolation Forest

Isolation Forest is a tree-based unsupervised anomaly-detection algorithm. It isolates anomalies by randomly splitting feature space; anomalies require fewer splits to isolate. It uses engineered tabular features such as RMS vibration, kurtosis, crest factor, dominant frequency, band energies, temperature, current, and their trends. It provides a simple, fast baseline against the deep-learning models.

11.3 LSTM Autoencoder

The LSTM Autoencoder is the primary unsupervised anomaly-detection model. It is trained only on healthy operating data, learns the machine's healthy temporal pattern, and flags abnormal windows when reconstruction error is high. LSTM autoencoders are widely used for vibration-based anomaly detection because they model sequential dependencies and reconstruct normal sensor patterns effectively. [web:54][web:61]

```
Input: sliding window (time steps, features)

Encoder:
LSTM(64, return_sequences=True)
LSTM(32)

Bottleneck:
RepeatVector(time steps)

Decoder:
LSTM(32, return_sequences=True)
LSTM(64, return_sequences=True)

Output:
TimeDistributed(Dense(features))
```

Training minimizes reconstruction error: L = (1/n) * sum((x_i - x_hat_i)^2). The anomaly score is the normalized reconstruction error; a threshold can be set using mean + 3 standard deviations, a percentile of healthy error, or the value maximizing F1-score.

11.4 1D-CNN Fault Classifier

The 1D-CNN performs supervised classification of fault types using vibration windows or spectral features. CNNs extract local patterns from vibration signals; STFT/CNN approaches have been applied to rotor-defect prediction. [web:51]

```
Input: (window length, channels)
```

```
Conv1D(32, kernel_size=3, relu) → BatchNorm → MaxPool
Conv1D(64, kernel_size=3, relu) → BatchNorm → MaxPool
Conv1D(128, kernel_size=3, relu) → GlobalAveragePooling
Dense(64, relu) → Dropout(0.3)
Dense(num_classes, softmax)
```

Class	Condition
0	Healthy
1	Bearing wear
2	Rotor imbalance
3	Misalignment
4	Overheating
5	Motor overload / abnormal current
6	Sensor anomaly

Use categorical cross-entropy loss and evaluate accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Recall is especially important because missing a developing fault can cause unplanned downtime.

11.5 CNN-LSTM RUL Model

The CNN-LSTM model estimates Remaining Useful Life from degradation sequences. The CNN extracts local signal patterns while the LSTM captures long-term temporal degradation. CNN-LSTM approaches are widely used for RUL prediction on C-MAPSS and related industrial prognostics datasets. [web:50][web:53]

```
Input: (sequence length, features)

Conv1D(64, kernel_size=3, relu) → MaxPool
Conv1D(128, kernel_size=3, relu) → MaxPool
LSTM(64, return_sequences=True)
LSTM(32)
Dense(32, relu) → Dropout(0.2)
Dense(1, linear) # RUL in hours or cycles
```

For C-MAPSS, RUL_t = RUL_max - t, commonly capped at 125–130 cycles. Evaluate using MAE, RMSE, R-squared, prediction-error distribution, and the percentage of predictions within a practical tolerance band.

12. MLOps Architecture

Tool	Function
DVC	Dataset and model versioning
MLflow	Experiment tracking and model registry
Evidently AI	Data drift and model-performance monitoring
Git / GitHub	Code version control
Docker	Reproducible environment
GitHub Actions	CI/CD automation
Amazon S3	Artifact and dataset storage

Workflow: code or data change → DVC tracks dataset/model version → training pipeline runs → MLflow logs parameters, metrics, artifacts → model evaluation → model registered if improved → deployed to inference endpoint → Evidently monitors live data → drift detected → retraining trigger activated.

Drift type	Description	Example
Data drift	Input distribution changes	Sensor baseline shifts after replacement
Concept drift	Input-target relationship changes	New operating condition changes fault pattern
Prediction drift	Model output distribution changes	Sudden increase in anomaly scores

Drift type	Description	Example
Performance drift	Model accuracy degrades	More false alarms or missed faults

13. Agentic AI Layer

The agentic layer converts model predictions into practical maintenance decisions using Diagnostic, Inventory, and Scheduler agents.

13.1 Diagnostic Agent

Interprets anomaly score and fault class, analyzes FFT features, retrieves relevant maintenance procedures using RAG, explains likely root cause, and recommends inspection steps.

```
{
"diagnosis": "High-probability bearing wear",
"confidence": 0.86,
"evidence": [
"RMS vibration increased by 48%",
"Kurtosis above healthy baseline",
"Dominant frequency shifted toward bearing-defect band",
"Temperature rising trend detected"
],
"recommended_action": "Inspect bearing and schedule replacement"
}
```

13.2 Inventory Agent

Identifies required spare parts, queries SAP MM for stock availability, checks procurement lead time, and determines whether a purchase requisition is needed.

```
{
"required_part": "6205-2RS Bearing",
"material_number": "100245",
"available_stock": 0,
"lead_time_days": 7,
"action": "CREATE_PURCHASE_REQUISITION"
}
```

13.3 Scheduler Agent

Assesses maintenance urgency, checks technician/work-center availability, determines priority, prepares the SAP PM work-order payload, and recommends a maintenance window.

14. SAP Integration Design

The AI engine integrates with SAP S/4HANA through SAP BTP, SAP Integration Suite, SAP API Management, SAP Destination Service, OAuth 2.0, and OData APIs. SAP provides Maintenance Notification APIs that support creating, reading, updating, and processing maintenance notifications through OData. [web:67][web:68]

14.1 SAP PM Workflow

```
AI anomaly detected
↓
Risk threshold exceeded
↓
Diagnostic Agent creates diagnosis
↓
Scheduler Agent determines priority
↓
SAP BTP prepares OData payload
↓
SAP PM Maintenance Notification created
↓
Maintenance work order created / linked
↓
Technician assigned and maintenance executed
```

14.2 SAP MM Workflow

```
Diagnostic Agent identifies spare part
↓
Inventory Agent calls SAP MM OData API
↓
Stock available?
■■■ Yes → Reserve/plan part for PM work order
■■■ No → Create purchase requisition
```

14.3 Example SAP PM Payload

```
{
"MaintenanceNotification": {
"MaintenanceNotificationType": "M1",
"MaintenanceNotificationText":
"High vibration anomaly detected on Conveyor Drive Motor",
"Equipment": "10000021",
"FunctionalLocation": "PLANT01-LINE2-MOTOR01",
"MaintPriority": "1",
"ReportedByUser": "AI_PREDICTIVE_ENGINE",
"MaintNotificationLongText":
"LSTM Autoencoder anomaly score 0.87. Predicted failure mode:
bearing wear. Estimated RUL: 38 hours."
}
}
```

14.4 Security Design

MQTT communication is secured with TLS; AWS IoT Core device certificates authenticate the ESP32; OAuth 2.0 secures SAP BTP-to-S/4HANA communication; SAP Destination Service stores credentials securely; role-based access protects Fiori users; API validation, rate limiting, and audit logs protect every AI-triggered SAP action.

15. SAP Fiori Dashboard

Page	Content
Asset Health Overview	Asset list, health score, status, critical alerts
Asset Detail	Live telemetry, trends, FFT plot, anomaly score, RUL
Alerts	Active alerts, severity, timestamp, recommended action
Maintenance Orders	SAP PM work orders and notifications
Inventory	Spare-part stock, lead time, requisition status
Model Insights	Model confidence, drift status, last retraining date
Reports	Downtime avoided, alerts generated, maintenance performance

Key widgets include live vibration waveform, FFT spectrum chart, temperature trend, current trend, asset-health gauge, RUL countdown, anomaly-score gauge, SAP work-order status, spare-part availability card, and maintenance-history timeline.

16. Technology Stack

Layer	Technologies
Hardware / Edge	ESP32, MPU-6050, DS18B20/DHT22, ACS712, DC motor, Wokwi
Cloud / Data	AWS IoT Core, AWS Lambda, Amazon Timestream, DynamoDB, S3, SNS, CloudWatch
AI / ML	Python, NumPy, Pandas, SciPy, Scikit-learn, TensorFlow/Keras or PyTorch, Matplotlib/Plotly
MLOps	DVC, MLflow, Evidently AI, Git, GitHub Actions, Docker, S3
Agentic AI	LangGraph or CrewAI, LLM, LangChain/LlamaIndex, FAISS/Chroma/Pinecone
SAP	SAP BTP, Integration Suite, API Management, Destination Service, OData, S/4HANA, PM, MM, Fiori/SAPUI5

17. Implementation Plan

Phase	Activities	Duration
1. Requirements & Design	Define asset, sensors, failure modes, KPIs, architecture, data model, APIs	1–2 weeks
2. Edge Device	Build ESP32 firmware, connect sensors, MQTT publishing, Wokwi fallback	2–3 weeks
3. AWS Pipeline	IoT Core, rules, Lambda, Timestream/DynamoDB, S3, telemetry testing	2–3 weeks
4. AI Models	Feature engineering, data collection/simulation, model training and evaluation	3–4 weeks
5. MLOps	DVC, MLflow, Evidently AI, registry, retraining workflow	1–2 weeks
6. Agentic AI	Diagnostic, Inventory, Scheduler agents, RAG	2–3 weeks
7. SAP Integration	BTP setup, destinations, OAuth, Integration Suite, PM/MM workflows	3–4 weeks
8. Fiori Dashboard	Asset health, telemetry charts, alerts, SAP actions, inventory	2–3 weeks
9. Testing & Demo	Unit, integration, SAP, model, end-to-end testing; report and presentation	2–3 weeks

18. Testing Strategy

Test case	Expected result
Normal motor operation	No anomaly alert; high health score
Gradual vibration increase	Anomaly score rises; watchlist alert
Sudden imbalance introduced	High anomaly score; SAP PM work order created
Temperature exceeds threshold	Warning or critical alert generated
Current spike detected	Possible overload diagnosis
Spare part available	SAP MM confirms stock; work order scheduled
Spare part unavailable	SAP MM purchase requisition triggered
SAP API failure	Retry/error log generated; alert sent
Sensor data missing	Data-quality alert raised
Model drift detected	Retraining workflow triggered

19. KPIs and Evaluation

Category	KPI	Target / purpose
Model	Accuracy, precision, recall, F1	Measure prediction quality
Model	False-positive rate	Reduce unnecessary alerts
RUL	MAE, RMSE, R-squared	Measure RUL prediction accuracy
System	End-to-end alert latency	Under 5–10 seconds
System	SAP work-order success rate	Above 95%
System	Dashboard refresh latency	Under 3 seconds
Business	Unplanned downtime reduction	Improved asset availability
Business	Spare-part stockout reduction	Better MM planning
Business	Emergency repair reduction	Lower maintenance cost

20. Risk Analysis

Risk	Impact	Mitigation
Insufficient fault data	Weak supervised model	Use simulation, fault injection, C-MAPSS, augmentation
Sensor noise	False anomalies	Apply filtering and robust thresholds
SAP sandbox limitations	Integration issues	Use mock OData services or simulated SAP endpoints

Risk	Impact	Mitigation
AWS free-tier limits	Service interruption	Optimize message frequency and use local buffering
Model drift	Poor predictions	Use Evidently AI and retraining
High false alarms	Engineer fatigue	Tune thresholds and use multi-signal confirmation
Hardware failure	Data gaps	Use Wokwi simulation and data replay
Security risk	Unauthorized access	Use TLS, certificates, OAuth, least-privilege access

21. Future Enhancements

• Extend the system to multiple assets and plants.

• Add edge inference using TensorFlow Lite Micro.

• Integrate with SCADA, PLC, and OPC-UA systems.

• Add generative AI-based maintenance-report generation.

• Use computer vision for visual inspection.

• Add reinforcement learning for maintenance scheduling.

• Implement predictive spare-part demand forecasting.

• Add blockchain-based maintenance audit trails.

• Use SAP Event Mesh for enterprise event streaming.

• Build a mobile Fiori app for technicians.

22. Conclusion

This project delivers a complete Industry 4.0 predictive-maintenance solution that connects physical equipment, cloud AI, agentic intelligence, and SAP S/4HANA. Its major strength is the closed-loop ERP action: sensor anomaly → AI diagnosis → RUL prediction → agent decision → SAP PM work order → SAP MM spare-part action → Fiori visibility.

By combining ESP32-based IoT sensing, AWS serverless architecture, FFT-driven feature engineering, LSTM Autoencoder anomaly detection, 1D-CNN fault classification, CNN-LSTM RUL prediction, MLOps, and SAP BTP integration, the project demonstrates how predictive maintenance can become an autonomous enterprise process rather than a passive analytics dashboard.